

Class 17

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Downstream Analysis

We need to install **tximport** package for straightforward import of Kallisto results.

- To troubleshoot import, presence of the quant folders were verified using the `dir()` command and the “rhdf5” bioconductor package was installed to read the **abundance.h5** files.

```
setwd("~/Desktop/class17_results")
dir()
```

```
[1] "SRR2156848_quant"  "SRR2156849_quant"  "SRR2156850_quant"
[4] "SRR2156851_quant"  "Untitled_files"     "Untitled.html"
[7] "Untitled.pdf"       "Untitled.qmd"       "Untitled.rmarkdown"
```

```
# BiocManager::install("tximport")
#BiocManager::install("rhdf5")
library(tximport)
library(rhdf5)

folders <- dir(pattern="SRR21568*")
samples <- sub("_quant", "", folders)
files <- file.path( folders, "abundance.h5" )
names(files) <- samples

txi.kallisto <- tximport(files, type = "kallisto", txOut = TRUE)
```

1 2 3 4

```
head(txi.kallisto$counts)
```

	SRR2156848	SRR2156849	SRR2156850	SRR2156851
ENST00000539570	0	0	0.00000	0
ENST00000576455	0	0	2.62037	0
ENST00000510508	0	0	0.00000	0
ENST00000474471	0	1	1.00000	0
ENST00000381700	0	0	0.00000	0
ENST00000445946	0	0	0.00000	0

How many transcripts do we have for each sample?

```
colSums(txi.kallisto$counts)
```

SRR2156848	SRR2156849	SRR2156850	SRR2156851
2563611	2600800	2372309	2111474

How many transcripts are detected in at least one sample?

```
sum(rowSums(txi.kallisto$counts)>0)
```

```
[1] 94561
```

We also have to filter out the annotated scripts with no reads and those with no change over the samples:

```
to.keep <- rowSums(txi.kallisto$counts) > 0
kset.nonzero <- txi.kallisto$counts[to.keep,]
keep2 <- apply(kset.nonzero,1,sd)>0
x <- kset.nonzero[keep2,]
```

Principal Component Analysis

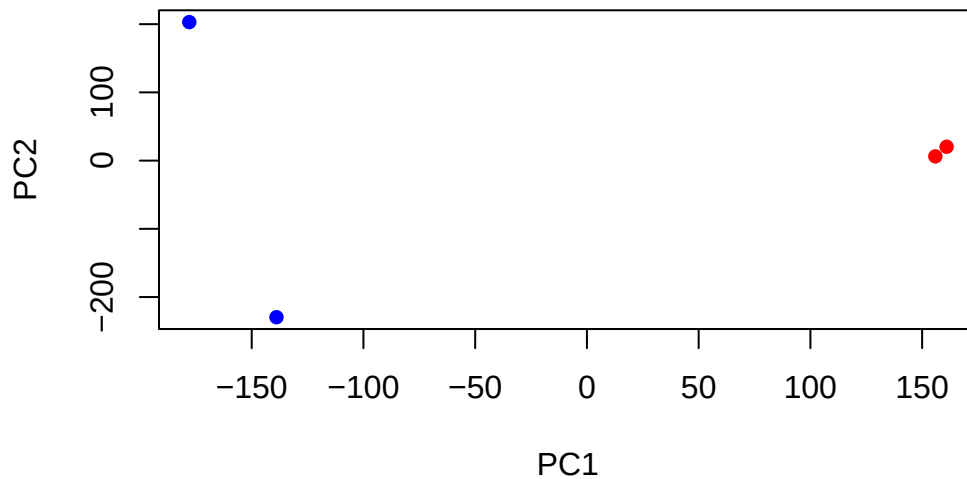
```
pca <- prcomp(t(x), scale=TRUE)
summary(pca)
```

Importance of components:

	PC1	PC2	PC3	PC4
Standard deviation	183.6379	177.3605	171.3020	1e+00
Proportion of Variance	0.3568	0.3328	0.3104	1e-05
Cumulative Proportion	0.3568	0.6895	1.0000	1e+00

Visualizing the summarized transcriptomic profiles of each sample

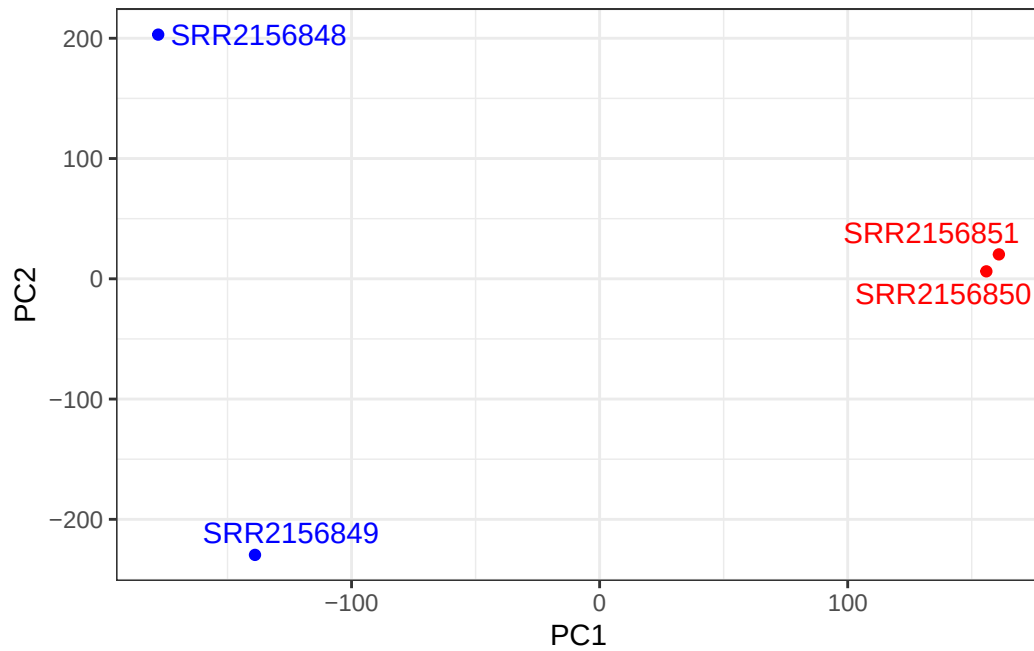
```
plot(pca$x[,1], pca$x[,2],  
     col=c("blue","blue","red","red"),  
     xlab="PC1", ylab="PC2", pch=16)
```



Q. Use ggplot to make a similar figure of PC1 vs PC2 and a separate figure PC1 vs PC3 and PC2 vs PC3.

- PC1 vs PC2

```
library(ggplot2)  
library(ggrepel)  
  
mycols <- c("blue","blue","red","red")  
  
ggplot(pca$x) +  
  aes(PC1, PC2, label=rownames(pca$x)) +  
  geom_point( col=mycols ) +  
  geom_text_repel( col=mycols ) +  
  theme_bw()
```

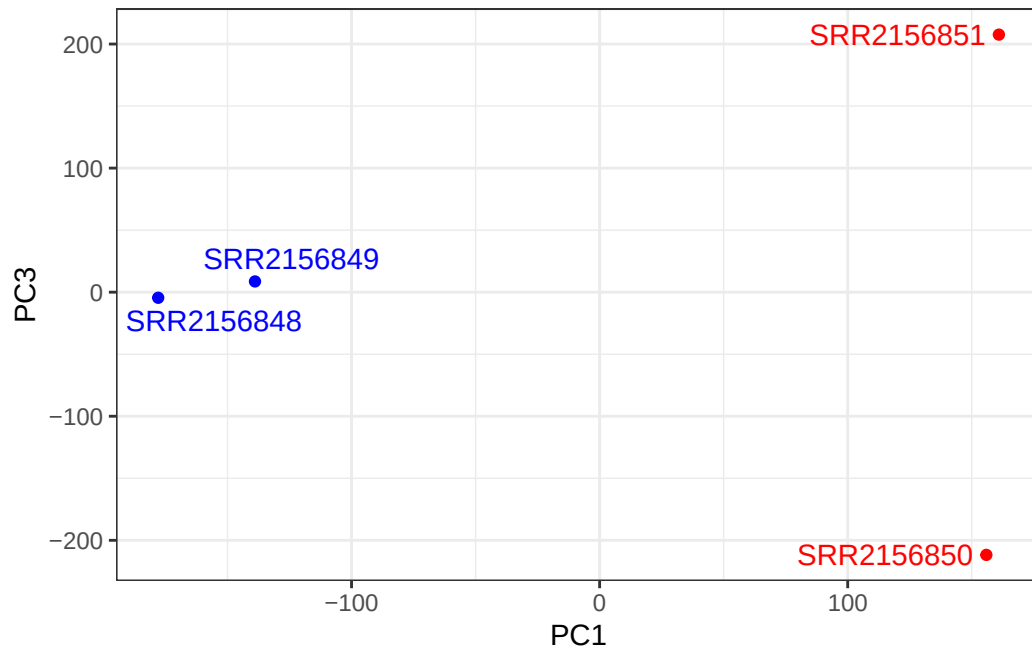


-PC1 vs PC3

```
library(ggplot2)
library(ggrepel)

mycols <- c("blue","blue","red","red")

ggplot(pca$x) +
  aes(PC1, PC3, label=rownames(pca$x)) +
  geom_point( col=mycols ) +
  geom_text_repel( col=mycols ) +
  theme_bw()
```



- PC2 vs PC3

```
library(ggplot2)
library(ggrepel)

mycols <- c("blue","blue","red","red")

ggplot(pca$x) +
  aes(PC2, PC3, label=rownames(pca$x)) +
  geom_point( col=mycols ) +
  geom_text_repel( col=mycols ) +
  theme_bw()
```

